

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – EXPERIMENTS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO3 – Precision (BL3)	Assessment:	Experiments
Weightage:	45%	Total Marks:	100
CO3 Statement:	Design processor organization by constructing pipelined datapath structures, identifying hazard types (data, control, structural), and comparing superscalar techniques such as out-of-order execution, speculative execution, and simultaneous multithreading (SMT).		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

- Identify the relevant concepts of register transfers, ALU operations, bus organization, pipelining, hazards, and superscalar processing.
- Analyze the execution of data transfers, arithmetic and logic operations, instruction processing, and parallel execution.
- Apply suitable techniques such as multiple buses, pipelining, stalling, forwarding, branch prediction, and speculative execution.
- Compute performance measures such as execution time, clock cycles, speedup, throughput, and resource utilization.
- Interpret the results by evaluating performance improvements, bottlenecks, and the suitability of each architectural technique.

QUESTIONS

1. Construct a pipeline scenario containing dependent instructions, branch instructions, and competing resource requirements, and apply techniques such as pipeline stalling, operand forwarding, branch prediction, and resource duplication to minimize performance losses.
2. Develop a computer system model using a hierarchical bus structure to transfer data among the CPU, memory, and I/O devices, and analyse how bus priority, bandwidth, and transfer conflicts affect overall system performance.
3. Develop a processor model that supports multiple instruction issue, out-of-order execution, and speculative execution, and analyse how instruction scheduling, execution units, and simultaneous multithreading affect instruction throughput and processor resource utilization.
4. Design and simulate a processor data path that performs register-to-register data transfers and arithmetic operations such as increment, decrement, addition, and subtraction, and analyse the control signals and execution sequence involved in each operation.
5. Simulate a pipelined processor executing a sequence of arithmetic, memory, and branch instructions, and calculate the pipeline execution time, throughput, and speedup by comparing the results with sequential instruction execution.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage(%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Marks Evaluation:

Q. No	Understanding of Concepts(25)	Experimental Design (30)	Use of Tools(20)	Analysis of Results (15)	Documentation (10)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

Course Coordinator

Faculty Incharge

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value	Flag
A	01	S 0
BC	05 05	Z 1
DE	03 00	AC 0
HL	00 00	P 1
PSW	00 00	C 0
PC	42 2F	
SP	FF FF	
Int-Reg	00	

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

0 - + 00

Update Memory

Load me at

```

1 MVI A, 05H
2 STA 8050H
3
4 MVI A, 03H
5 STA 8051H
6
7 LDA 8050H
8 MOV B, A
9
10 LDA 8051H
11 ADD B
12 STA 8052H
13
14 LDA 8052H
15 INR A
16 STA 8053H
17
18 CPI 09H
19 JZ TAKEN
20
21 MVI A, 00H
22 STA 8054H
23 JMP STOP
24
25 TAKEN: MVI A, 01H
26 STA 8054H
27
28 STOP: HLT
29

```

Start: 8050

Address (Hex)	Address	Data
1F72	8050	5
1F73	8051	3
1F74	8052	8
1F75	8053	9
1F76	8054	1
1F77	8055	0
1F78	8056	0
1F79	8057	0
1F7A	8058	0
1F7B	8059	0
1F7C	8060	0
1F7D	8061	0
1F7E	8062	0
1F7F	8063	0

Line No. Assembler Message

0 Program assembled successfully

Simulator: Idle

33°C Mostly sunny

Search

ENG IN

11:11 12.08.2026

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value	Flag
A	01	S 0
BC	05 00	Z 1
DE	00 00	AC 0
HL	00 00	P 1
PSW	00 00	C 0
PC	42 2F	
SP	FF FF	
Int-Reg	00	

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

0 - + 00

Update Memory

Load me at

```

1 ; Pipeline Hazard Demonstration
2 ; Memory: 8050H - 805FH
3
4 MVI A, 05H
5 STA 8050H
6
7 MVI A, 03H
8 STA 8051H
9
10 LDA 8050H
11 MOV B, A
12
13 LDA 8051H
14 ADD B
15 STA 8052H
16
17 LDA 8052H
18 INR A
19 STA 8053H
20
21 CPI 09H
22 JZ NEXT
23
24 MVI A, 00H
25 STA 8054H
26 JMP STOP
27
28 NEXT: MVI A, 01H
29 STA 8054H
30
31 STOP: HLT

```

Start: 8085

Address (Hex)	Address	Data
1F95	8085	5
1F96	8086	3
1F97	8087	8
1F98	8088	9
1F99	8089	1
1F9A	8090	0
1F9B	8091	0
1F9C	8092	0
1F9D	8093	0
1F9E	8094	0
1F9F	8095	0
1FA0	8096	0
1FA1	8097	0
1FA2	8098	0

Line No. Assembler Message

0 Program assembled successfully

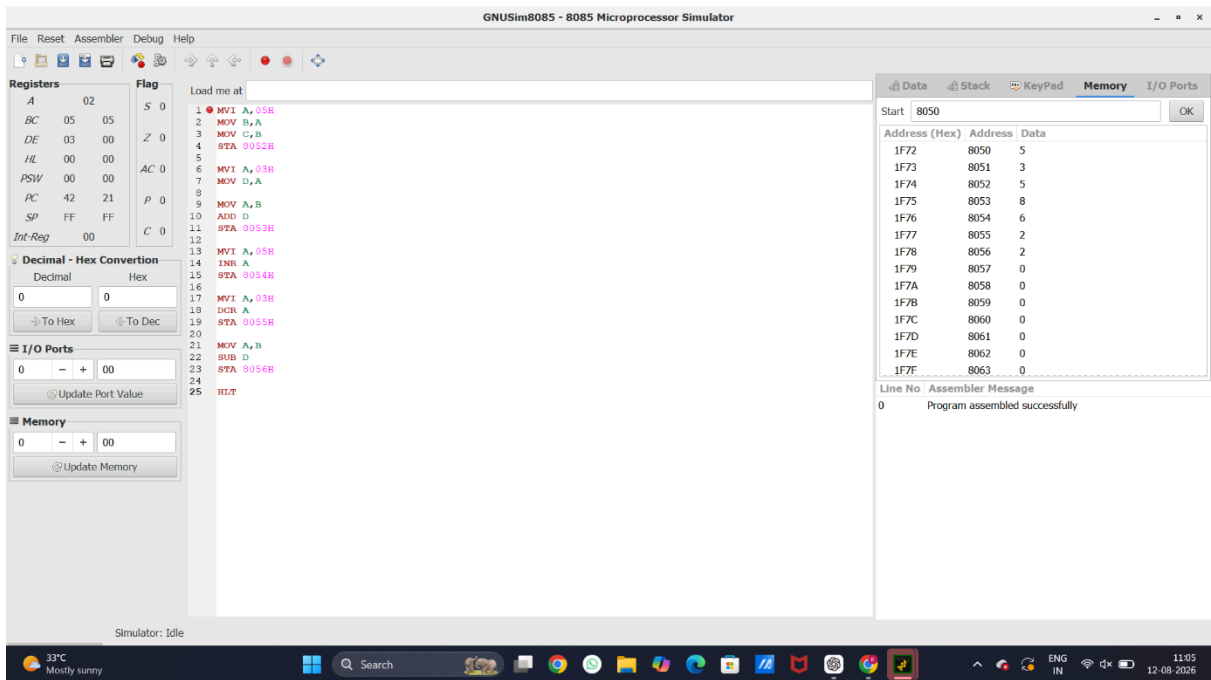
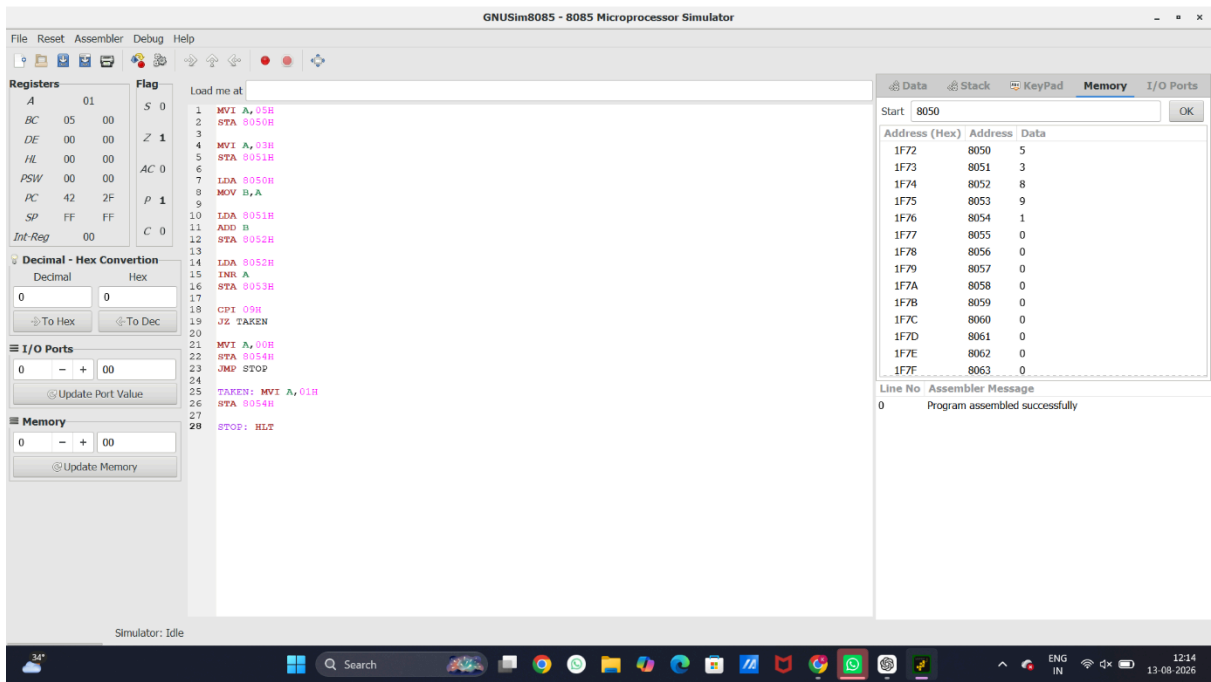
Simulator: Idle

32°C Sunny

Search

ENG IN

10:48 12.08.2026



FileResetAssemblerDebugHelp

Registers

A	01	S	0
BC	05 00	Z	1
DE	00 00	AC	0
HL	00 00	P	1
PSW	00 00	C	0
PC	42 2F		
SP	FF FF		
Int-Reg	00		

Flag

S	0
Z	1
AC	0
P	1
C	0

Decimal - Hex Conversion

Decimal

Hex

0

0

To Hex

To Dec

I/O Ports

0

-

+

00

Update Port Value

Memory

0

-

+

00

Update Memory

Load me at

1

MVI A, 05H

2

STA 8050H

3

4

MVI A, 03H

5

STA 8051H

6

7

LDA 8050H

8

MOV B, A

9

10

LDA 8051H

11

ADD B

12

STA 8052H

13

14

LDA 8052H

15

INR A

16

STA 8053H

17

18

CPI 00H

19

JZ TAKEN

20

21

MVI A, 00H

22

STA 8054H

23

JMP STOP

24

25

TAKEN: MVI A, 01H

26

STA 8054H

27

28

STOP: HLT

DataStackKeypadMemoryI/O Ports

Start

8050

OK

Address (Hex)

Address

Data

1F72

8050

8

1F73

8051

9

1F74

8052

5

1F75

8053

3

1F76

8054

8

1F77

8055

9

1F78

8056

0

1F79

8057

0

1F7A

8058

0

1F7B

8059

0

1F7C

8060

0

1F7D

8061

0

1F7E

8062

0

1F7F

8063

0

Line No

Assembler Message

0

Program assembled successfully

Simulator: Idle

35°

Search

ENG IN

12:32 13-08-2026